Narcolepsy — Scientific Due Diligence

Pharma investment committee brief · 2026-05-17 · n=3 deeply-analysed papers from 30-paper corpus

> ***Invest in autonomic-metabolic phenotyping of confirmed orexin-deficient narcolepsy type 1 using wearable-derived pulse rate/motor activity as the objective primary endpoint, targeting a sympathetic tone dysregulation mechanism that is biologically grounded, under-studied, and measurable with pre-specified device preprocessing protocols.***

Executive Summary

The narcolepsy therapeutic corpus analyzed (n=30 papers, 3 deep extractions) presents a field in early mechanistic consolidation with a critical evidentiary deficit: zero findings rise above 'speculative' certainty, and all 12 calibrated consensus items carry low extraction confidence (0.78–0.82). The evidence base is dominated by reviews, expert opinion, and small cohort studies, with no GRADE Moderate or High evidence present in the deep-extracted set. The highest-signal mechanistic opportunity is autonomic-metabolic dysregulation downstream of orexin deficiency, identified in 3 phenotype extractions and supported by the only prospective wearable dataset in the corpus (PMC13175547). This mechanism is under-published relative to its biological plausibility and offers an objective, device-derived endpoint (wearable pulse rate/motor activity coupling) that bypasses the field's dominant reliance on subjective scales. The single methodological insight that must drive Phase II protocol design: preprocessing and validity thresholds for wearable biomarkers must be pre-specified and cross-device validated before efficacy measurement begins — PMC13175547 demonstrates that ~23.5% of epoch-level wearable data is invalid without rigorous artifact exclusion. The deal-breaker risk is the absence of any clean baseline stratified by prior treatment, comorbid psychopathology, or orexin-deficiency confirmation status; mitigation requires prospective CSF orexin-A measurement as an inclusion stratifier and wearable run-in period of ≥14 days prior to randomization.

Committee Briefing — 60 Seconds

- SIGNAL: The entire 30-paper narcolepsy corpus yields zero findings above 'speculative' certainty — but one prospective wearable study (n=110, CITE:PMC13175547) provides the field's only objective physiological endpoint data, making autonomic dysregulation the only investment-grade mechanistic target in this evidence set.
- BET: Narcolepsy Type 1 autonomic-metabolic phenotype — sympathetic tone dysregulation measurable via wearable pulse rate/motor activity decoupling — is biologically grounded in orexin-locus coeruleus circuitry, under-published (n=3 phenotype mentions vs. its mechanistic centrality), and supported by device-derived data with R^2c 0.76–0.80 [1].
- AVOID: The modafinil-psychosis case series [2] and the Avadel-sponsored Delphi [3] are both GRADE Very Low and methodologically disqualified as Phase II hypothesis generators — treating either as evidence would expose the program to immediate SAB challenge.

- CRITICAL PROTOCOL GATE: Pre-specify wearable artifact exclusion thresholds before randomization, mandate a 14-day run-in for individual baseline capture, and cross-validate the preprocessing pipeline on ≥1 additional device — without this, the primary endpoint is unpublishable and the program undefendable [1].
- ENROLLMENT ANCHOR: Stratify exclusively on CSF orexin-A ≤110 pg/mL AND HLA-DQB1*06:02 positivity — no paper in this corpus does this, meaning a biomarker-stratified Phase II would generate genuinely novel data and a defensible precision medicine narrative.
- COMPETITIVE MOAT: The ON-SXB/LUMRYZ commercial launch [3] means the program must not compete on symptomatic efficacy alone — the differentiated thesis is biomarker-guided patient selection and objective autonomic endpoint, positioning the program for companion diagnostic co-development and label differentiation.
- DEAL-BREAKER TO RESOLVE FIRST: Commission a simulation study using PMC13175547 raw data to estimate fixed-effect R^2m independently of random effects — if the population-level autonomic signal is small, the primary endpoint fails before Phase II begins, and the 20M€ allocation must pivot to autoimmune biomarker validation using external literature.

1. Target Trial Emulation Landscape

The corpus contains zero formal target trial emulations using g-methods, marginal structural models, or propensity-scored causal inference frameworks. The closest approximation is the Avadel-sponsored modified Delphi on once-nightly sodium oxybate (ON-SXB) management (PMC13172060), which generates prescribing consensus rather than causal effect estimates and is disqualified from TTE status by its industry-directed question framing, n=7 panel, and 67% consensus threshold. The modafinil-psychosis systematic review (PMC13167382) employs a numerator-only case aggregation design with no denominator, making incidence estimation impossible and Phase II hypothesis generation from this source invalid. No re-purposing signals for drugs outside the orexin/sodium oxybate class (e.g., IL-6 antagonists, low-dose naltrexone, metformin) are present in any of the 30 papers. The TTE landscape for narcolepsy is essentially empty, representing both a gap and an opportunity: a well-designed observational emulation of ON-SXB versus modafinil on autonomic endpoints would be a de novo contribution.

DOI	Drug class	Design quality	Phase II signal	Note
[3]	Sodium oxybate (once-nightly formulation, ON-SXB/LUMRYZ)	weak	equivocal	Industry-sponsored Delphi (n=7, Avadel-curated panel) generates prescribing recommendations for ON-SXB but lacks causal inference architecture, denominator data, or patient outcome validation — useful only for dosing convention reference, not Phase II hypothesis generation (PMC13172060).
[2]	Modafinil (wake-promoting agent, dopamine transporter blocker)	weak	argues_against	Numerator-only case series aggregate (n=24 psychosis events) cannot estimate true modafinil-psychosis incidence, but mechanistically flags hyperdopaminergia and orexinergic-GABAergic interaction as safety signals requiring prospective monitoring in any stimulant-inclusive Phase II arm (PMC13167382).

DOI	Drug class	Design quality	Phase II signal	Note
[1]	No drug — wearable biomarker preprocessing framework	adequate	supports	Prospective cohort (n=110 NT1 patients) establishes wearable pulse rate/motor activity coupling as a feasible objective endpoint with defined preprocessing thresholds, directly informing Phase II endpoint operationalization for autonomic phenotype trials (PMC13175547).

2. Objective Biomarker Bridges (Candidate Phase II Endpoints)

The narcolepsy corpus is critically immature with respect to objective, validated biomarkers suitable for Phase II primary endpoints. The dominant measurement paradigm across the 30-paper corpus is clinician-rated or self-reported symptom scales (Epworth Sleepiness Scale mean 18.5, weekly cataplexy rate mean 22.2), both flagged as speculative-certainty findings (PMC13172060). The single paper employing an objective physiological measurement approach is PMC13175547, which demonstrates that wearable-derived pulse rate and motor activity can be systematically captured in NT1 patients with a defined preprocessing pipeline — ~23.5% epoch invalidity rate and a ≥70% valid-data-per-day threshold. Critically, this study reports a mixed-model R^2c of 0.76–0.80 for pulse rate/motor activity association, but does not disaggregate the fixed-effect R^2m, obscuring how much variance is explainable at the population versus individual level. No cytokine panels, neurofilament light chain measurements, autoantibody assays, complement markers, or CPET-derived cardiopulmonary reserve data appear in any of the 30 papers. The biomarker bridge landscape is narrow but not empty: wearable autonomic coupling is the only candidate with prospective data.

Biomarker	n papers	Correlation	Exemplars	Phase II readiness	Note
Wearable-derived pulse rate / motor activity coupling (epoch-level, ≥70% valid day threshold)	1	moderate	[1]	with_validation	R^2c 0.76–0.80 for pulse rate/motor activity association in NT1 (n=110); requires cross-device validation and fixed-effect R^2m disaggregation before use as primary endpoint — preprocessing protocol is the trial's most critical pre-specified deliverable (PMC13175547).
Weekly cataplexy rate (clinician-confirmed, video-polysomnography anchor)	1	weak	[3]	with_validation	Mean weekly cataplexy rate 22.2 reported in the ON-SXB Delphi context (PMC13172060), but this is expert-opinion-derived, not prospectively measured; requires PSG-anchored confirmation and inter-rater reliability data before deployment as a Phase II endpoint.

Biomarker	n papers	Correlation	Exemplars	Phase II readiness	Note
Epworth Sleepiness Scale (ESS) score	1	weak	[3]	no	ESS mean 18.5 cited in the industry Delphi (PMC13172060) but is a subjective self-report instrument with well-documented ceiling effects in severe NT1 — unsuitable as a Phase II primary endpoint without an objective anchor; retain as exploratory secondary only.
CSF orexin-A (hypocretin-1) concentration	1	moderate	[3]	with_validation	Orexin deficiency is cited as the established NT1 pathophysiology (PMC13172060), but no paper in the corpus prospectively measures CSF orexin-A as a stratification or efficacy biomarker — critical gap; recommend as an inclusion criterion stratifier rather than a dynamic endpoint.

3. Clean Baseline Subset (Variant + Vaccination Stratified)

Size: 0 studies (0% of corpus)

The 'clean baseline' analytical lens, designed to filter for vaccination-stratified and variant-era-controlled studies, yields zero qualifying papers from this narcolepsy corpus. This is not a gap created by poor reporting — it reflects the disease context: narcolepsy type 1 is primarily understood through the orexin-autoimmunity paradigm precipitated by H1N1 infection or vaccination, and none of the 30 papers stratify patients by immunological exposure history. The implication for protocol design is significant: the committee cannot rely on any existing study to establish a variant-clean or immunologically-characterized baseline population. Phase II enrollment must therefore prospectively collect vaccination history, prior infectious exposures, and — critically —

HLA-DQB1*06:02 haplotype status, which is present in >90% of NT1 cases and constitutes the only currently available biological stratification anchor. Absence of this data in the corpus is a protocol design gap, not a scientific dead end.

Key findings from this clean subset:

- No paper in the 30-paper corpus stratifies by vaccination dose or variant era — the 'clean baseline' subset does not exist in this narcolepsy corpus [2].
- The bias audit records zero instances of variant/vaccine confounding adjustment across all reviewed papers — this is a structural absence, not a reporting omission [1].
- Variant/vaccination stratification is not applicable to classic narcolepsy type 1 etiology in the same sense as post-COVID syndromes; however, the H1N1 vaccination-narcolepsy signal (historical, not represented in this corpus) represents the closest analogue and is entirely absent from the evidence base.

4. Mechanistic Opportunity Map (Blue vs. Red Ocean)

The 20M€ should NOT go toward dopaminergic-modafinil safety characterization — this is a red ocean of methodologically weak case reports generating unquantifiable risk signals with no therapeutic target [2]. It should NOT go toward autoimmunity target validation without first expanding beyond this 30-paper corpus, as the autoimmune evidence here is entirely referenced rather than measured. The allocation should concentrate on autonomic-metabolic phenotyping: this mechanism sits at the intersection of established orexin biology, a prospective wearable dataset [1], and a complete absence of competitor Phase II programs using objective autonomic endpoints. A Phase II program anchored on wearable-derived pulse rate/motor activity coupling in HLA-DQB1*06:02-confirmed, CSF-orexin-stratified NT1 patients would be scientifically differentiated, operationally feasible, and positioned at the blue ocean intersection of under-studied mechanism and emerging measurement technology. The vascular-endothelial space is noted as a speculative second frontier requiring external literature support.

Mechanism	n papers	Mean GRADE	Signal consistency	Ocean	Exemplars
Autonomic Metabolic	3	Low	moderate	BLUE	[1]
Autoimmunity	1	Low	moderate	NEUTRAL	[3]
Other	2	Low	low	RED	[2]
Vascular Endothelial	0	Low	low	BLUE	
Viral Reservoir	0	Low	low	NEUTRAL	

Autonomic Metabolic — Sympathetic tone dysregulation secondary to orexin deficiency is identified in 3 phenotype extractions and is the only mechanism supported by prospective objective data (wearable pulse rate/motor activity, R^2c 0.76–0.80, n=110) [1]. The mechanism is biologically coherent — orexin projections directly modulate locus coeruleus noradrenergic output — but is dramatically under-published relative to its signal strength, creating a genuine first-mover opportunity for a Phase II program anchored on autonomic biomarkers.

Autoimmunity — Autoimmunity is referenced as the established NT1 pathophysiology via orexin neuron loss in the Delphi paper [3], but no paper in the corpus prospectively measures autoimmune biomarkers (anti-TRIB2 antibodies, complement activation, T-cell clonotypes). The mechanism is well-accepted but evidentially thin in this corpus — investment here requires external literature synthesis beyond the 30-paper set.

Other — Dopaminergic and GABAergic mechanisms related to modafinil-induced psychosis are flagged [2] but represent an adverse event signal rather than a therapeutic target. The numerator-only case series design precludes any reliable effect size estimation, and the mechanism is clinically relevant only as a safety exclusion criterion for stimulant-class comparators in Phase II.

Vascular Endothelial — Vascular-endothelial mechanisms are entirely absent from this narcolepsy corpus. Given emerging evidence in related hypersomnia conditions linking endothelial dysfunction to sleep fragmentation, this represents a speculative but genuine blue ocean — however, the committee should be aware that this recommendation has no direct corpus support and would require external literature justification before allocation.

Viral Reservoir — Viral reservoir mechanisms are absent from the corpus. The H1N1-narcolepsy trigger is historically established but no paper in this set addresses persistent viral antigen burden or reactivation dynamics. This is a known gap rather than an investment opportunity without further corpus expansion.

5. Methodological Risk Index per Mechanism

Mechanism	GRADE Low+	Surveillance bias	No baseline	No vax/variant adj	Risk score
Autonomic Metabolic	100%	33%	67%	100%	**Moderate**
Autoimmunity	100%	0%	100%	100%	**High**
Other (Dopaminergic/gabaergic — Modafinil Safety)	100%	100%	100%	100%	**Critical**
Vascular Endothelial	100%	0%	100%	100%	**Critical**

Interpretations:

- *Autonomic Metabolic*: Despite being the best-evidenced mechanism in this corpus, the autonomic-metabolic phenotype carries moderate protocol risk due to device-specific preprocessing thresholds that are not yet cross-validated and a fixed-effect R^2m that is unreported — Phase II must pre-specify artifact exclusion rules and validate across ≥2 wearable platforms before locking the primary endpoint (CITE:PMC13175547).

- *Autoimmunity*: Autoimmunity is referenced as established NT1 pathophysiology (CITE:PMC13172060) but has zero prospective biomarker measurement in this corpus, making any Phase II program targeting this mechanism entirely reliant on external evidence and therefore high-risk for protocol hypothesis failure.
- *Other (Dopaminergic/gabaergic — Modafinil Safety)*: The modafinil-psychosis signal is based entirely on numerator-only case aggregation with extreme dose heterogeneity (100–12,000 mg/day) and publication bias — it is critically unsuitable as a Phase II hypothesis generator and should be treated exclusively as a safety monitoring signal (CITE:PMC13167382).
- *Vascular Endothelial*: Zero papers in the corpus address vascular-endothelial mechanisms in narcolepsy — any Phase II protocol anchored here would be building on a foundation entirely external to this evidence base, representing the highest possible methodological risk for this allocation.

6. Contradictions Matrix — Where Field Wisdom Falls Short

Sex distribution in narcolepsy-associated adverse events

Received wisdom: Narcolepsy and autoimmune sleep disorders skew female, consistent with broader autoimmune epidemiology.

Contradicting evidence: In the modafinil-psychosis systematic review (n=24), the male:female ratio was 13:11 — near-parity, with a slight male predominance. While the sample is small and the mechanism is adverse-event-specific rather than disease-intrinsic, it challenges the assumption that female-predominant enrollment is required for narcolepsy therapeutic trials targeting neuropsychiatric comorbidities [2]. *Sources:* [2] **Implication for Phase II design:** Phase II inclusion criteria should not impose sex-ratio quotas based on assumed female predominance; enroll to biological stratifiers (HLA haplotype, CSF orexin-A level) rather than demographic proxies — failure to do so risks systematic under-enrollment of male patients with psychosis-vulnerable profiles who may be the highest-need population for safety-monitored stimulant alternatives.

Subjective sleepiness as the dominant and sufficient primary endpoint

Received wisdom: ESS and clinician-rated cataplexy frequency are the field-standard primary endpoints for narcolepsy trials, accepted by regulators and reflected in all major approved trials.

Contradicting evidence: The wearable prospective cohort (n=110 NT1) demonstrates that motor activity and pulse rate provide objective, continuously sampled physiological signal with R^2c 0.76–0.80 — and that subjective scales are not even collected in this dataset as the comparator endpoint. The implicit argument is that autonomic physiology captures a disease dimension orthogonal to sleepiness ratings [1]. *Sources:* [1] **Implication for Phase II design:** Position wearable-derived autonomic coupling as the primary endpoint with ESS as a powered secondary — this inverts the conventional hierarchy, differentiates the trial scientifically, and provides continuous real-world data collection that supports a potential digital biomarker regulatory submission in parallel with the therapeutic NDA.

Industry consensus as a reliable prescribing evidence base

Received wisdom: Delphi expert consensus panels provide actionable clinical guidance when RCT evidence is absent, particularly for dosing and titration decisions.

Contradicting evidence: The ON-SXB Delphi (PMC13172060) was convened by the manufacturer (Avadel), panel questions were reviewed by the sponsor for 'consistency with prescribing information,' the panel comprised only 7 clinicians (below the 15–30 minimum recommended for Delphi validity), and no individual COI disclosures were published. Consensus threshold of 67% with n=7 means 5 votes constitute a clinical recommendation — this is insufficient to treat as independent evidence and could be challenged as promotional material at SAB review [3]. *Sources:* [3] **Implication for Phase II design:** Do not cite the ON-SXB Delphi as independent evidence for comparator selection or dosing benchmarks in Phase II protocol documents — it will be identified by a rigorous SAB reviewer as sponsor-directed opinion; instead, anchor dosing decisions to the approved prescribing information and independently published pharmacokinetic data.

Age profile of narcolepsy therapeutic targets

Received wisdom: Narcolepsy type 1 typically presents in adolescence/young adulthood; therapeutic trials target a relatively young population.

Contradicting evidence: In the modafinil-psychosis case series, mean age was 37.4 years (range 13–79), suggesting that the narcolepsy population experiencing clinically significant neuropsychiatric adverse events under current treatment is substantially older than the disease-onset profile — and that middle-aged patients represent an under-characterized safety-relevant subgroup [2]. *Sources:* [2] **Implication for Phase II design:** Stratify Phase II enrollment by age decade (≤25 / 26–45 / >45) to detect differential autonomic biomarker trajectories and adverse event rates — treating narcolepsy as a uniformly young-patient disease will generate a biologically heterogeneous trial population and obscure age-dependent treatment response signals.

7. Recommended Target Phenotype & Phase II Skeleton

Target phenotype: Narcolepsy Type 1 with objective autonomic dysregulation — sympathetic tone dysregulation measurable via wearable pulse rate/motor activity decoupling in CSF-orexin-confirmed, HLA-DQB1*06:02-positive patients

The autonomic-metabolic phenotype is the single mechanism in this corpus supported by prospective physiological data [1], biologically grounded in the established orexin-locus coeruleus-sympathetic axis, and measured by an objective endpoint technology (wearable photoplethysmography plus accelerometry) that bypasses the corpus-dominant problem of subjective self-report bias. The PMC13175547 dataset (n=110 NT1, prospective, clinical trial baseline) demonstrates that pulse rate and motor activity coupling can be captured with defined preprocessing pipelines — 23.5% epoch invalidity is manageable with a ≥70% valid-day threshold — and explains 76–80% of variance in mixed models. Critically, the fixed-effect-only R^2m is not reported, meaning individual-level random effects may account for substantial unexplained variance; this is the primary technical risk and must be addressed in a 14-day wearable run-in period before randomization to establish individual baselines. Patient stratification must be anchored on CSF orexin-A ≤110 pg/mL (NT1 diagnostic criterion) and HLA-DQB1*06:02 positivity to eliminate NT2 and idiopathic hypersomnia contamination — a failure mode not addressed in any of the 30 papers. The Avadel Delphi [3] confirms that current clinical management of NT1 is driven by provider opinion rather than biomarker-guided algorithms, creating a white space for a biomarker-stratified Phase II to generate genuinely differentiated data. The modafinil safety signal [2] informs comparator arm design: any modafinil-inclusive arm requires prospective neuropsychiatric monitoring with structured psychosis screening at each visit, given the documented hyperdopaminergia risk.

Phase II Design Skeleton

Inclusion criteria:

- Confirmed NT1: CSF orexin-A ≤110 pg/mL AND HLA-DQB1*06:02 positive
- Age 18–65 years (stratified by decade: 18–25 / 26–45 / 46–65)
- Epworth Sleepiness Scale ≥16 at screening (to ensure active disease burden)
- Willing and able to wear Empatica EmbracePlus continuously for ≥14-day run-in plus trial duration
- ≥70% valid wearable data days during 14-day run-in period (pre-randomization quality gate)
- Stable or no pharmacological narcolepsy treatment for ≥4 weeks prior to screening (washout standardization)

Exclusion criteria:

- Narcolepsy Type 2 or idiopathic hypersomnia (CSF orexin-A >110 pg/mL or HLA-DQB1*06:02 negative)
- Prior or current psychotic disorder (DSM-5), given modafinil-psychosis safety signal (CITE:PMC13167382)
- Concurrent use of dopamine agonists or antipsychotics that would confound autonomic biomarker interpretation
- Severe obstructive sleep apnea (AHI >30 events/hour) as a confounding autonomic signal source
- Implanted cardiac devices or conditions precluding valid photoplethysmography capture
- Pregnancy or planned pregnancy (physiological autonomic changes confound the primary endpoint)
- Inability to provide valid ≥70% wearable data coverage during run-in (device non-compliers excluded before randomization)

Primary endpoint: Change from baseline in wearable-derived pulse rate/motor activity coupling coefficient (mixed-effects model, fixed-effect estimate only, R^2m) at 12 weeks, using pre-specified Empatica EmbracePlus preprocessing pipeline with artifact thresholds ≥220 bpm / <40 bpm with >30 activity counts exclusion criteria (CITE:PMC13175547)

Secondary endpoints:

- Change in weekly clinician-confirmed cataplexy rate (video-PSG anchor at baseline and week 12)
- Epworth Sleepiness Scale total score change from baseline (subjective comparator to primary objective endpoint)
- Neuropsychiatric Safety Monitoring: Columbia Suicide Severity Rating Scale and PANSS-6 at every visit (CITE:PMC13167382)

- Proportion of participants achieving ≥3-point ESS reduction AND ≥20% improvement in pulse rate/motor activity coupling (composite responder analysis)
- Device adherence rate (% of randomized participants achieving ≥70% valid days per week throughout trial)

Comparator: Placebo (double-blind), with optional open-label extension to ON-SXB at week 12 for patients not achieving primary endpoint response — avoids the methodological limitations of the industry Delphi (CITE:PMC13172060) as a comparator justification anchor

Sample-size basis: Sample size cannot be reliably estimated from this corpus because fixed-effect R^2m for the wearable primary endpoint is not reported (CITE:PMC13175547) — the 14-day run-in period must generate individual-level variance estimates to power the study; a conservative minimum of n=120 per arm is recommended based on the 0.76–0.80 R^2c observed in the 110-patient feasibility cohort, pending a pre-protocol simulation study.

Deal-breakers

- Fixed-effect R^2m for wearable pulse rate/motor activity coupling is not reported — if population-level (fixed) effect is small and variance is primarily individual-level random effect, the biomarker cannot serve as a sensitive primary endpoint; this must be resolved in a pre-Phase II simulation using the PMC13175547 raw dataset before committing to the endpoint (CITE:PMC13175547).
- Cross-device validation of preprocessing thresholds is absent — the entire endpoint framework is defined on Empatica EmbracePlus data only; if regulatory or operational pressures require device substitution, the preprocessing pipeline validity collapses entirely (CITE:PMC13175547).
- CSF orexin-A measurement as an inclusion criterion will reduce enrollment rate substantially (lumbar puncture acceptance ~40–60% in ambulatory narcolepsy populations) and increase screening costs — protocol must budget ≥2x the expected randomization-to-screen ratio.
- The ON-SXB competitive landscape (LUMRYZ approved, sodium oxybate generics entering market) means the 20M€ program must demonstrate autonomic biomarker-guided patient stratification adds clinical utility beyond existing approved agents — failure to pre-specify a responder subgroup analysis by baseline autonomic coupling severity will leave the program without a differentiated commercial thesis (CITE:PMC13172060).
- Any psychosis event in a modafinil-containing comparator arm will trigger regulatory scrutiny given the published safety signal — the numerator-only case series (CITE:PMC13167382) precludes incidence quantification but guarantees that the SAB will demand a pharmacovigilance plan for dopaminergic agents in any arm of the trial.

References

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sodium oxybate for narcolepsy: a propensity score-matched cohort study Treatment of central disorders of hypersomnolence: an American Academy of Sleep Medicine clinical practice guideline Drug class utilization and age-related trends in narcolepsy and idiopathic hypersomnia: a claims database analysis Evidence of accidental dosing errors with immediate-release sodium oxybate: data from the US Food and Drug Administration Adverse Event Reporting System Accidental calcium, magnesium, potassium and sodium oxybates (Xywave) overdose: mistiming of a single night's narcolepsy medication leading to respiratory failure requiring mechanical ventilation Efficacy and safety of calcium, magnesium, potassium, and sodium oxybates (lower-sodium oxybate [LXB]; JZP-258) in a placebo-controlled, double-blind, randomized withdrawal study in adults with narcolepsy with cataplexy Once-nightly sodium oxybate (FT218) demonstrated improvement of symptoms in a phase 3 randomized clinical trial in patients with narcolepsy Effect of FT218, a once-nightly sodium oxybate formulation, on disrupted nighttime sleep in patients with narcolepsy: results from the randomized phase III REST-ON trial RESTORE: once-nightly oxybate dosing preference and nocturnal experience with twice-nightly oxybates Real-world experience of once-nightly sodium oxybate treatment in people with narcolepsy: Interim results from REFRESH The Delphi technique as a forecasting tool: issues and analysis An experimental application of the Delphi method to the use of experts Disrupted nighttime sleep in narcolepsy The Federal Controlled Substances Act: schedules and pharmacy registration Adherence to wakefulness promoting medication in patients with narcolepsy Clinical and practical considerations in the pharmacologic management of narcolepsy Understanding the patient experience with twice-nightly sodium oxybate therapy for narcolepsy: a social listening experiment Safety and efficacy of long-term use of sodium oxybate for narcolepsy with cataplexy in routine clinical practice Sleep disturbance is associated with cardiovascular and metabolic disorders Narcolepsy and cardiovascular health: a big picture perspective Life’s Essential 8: updating and enhancing the American Heart Association’s construct of cardiovascular health: a presidential advisory from the American Heart Association Practice guideline recommendations summary: disease-modifying therapies for adults with multiple sclerosis: report of the Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology OnabotulinumtoxinA for lower limb spasticity: guidance from a Delphi panel approach Dosing optimization of low-sodium oxybate in narcolepsy and idiopathic hypersomnia in adults: consensus recommendations Narcolepsy severity scale: a reliable tool assessing symptom severity and consequences *Neurology and Therapy*. 2026. PMC13172060